

JUNBAO LIANG

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SUMMARY

PhD candidate in Computational Chemistry, building scientific ML models for protein structure analysis. Developed a binding pocket prediction pipeline from protein structures using feature engineering and XGBoost classification, with scalable inference via spatial indexing. Experienced with MD workflows (AMOEBA/CHARMM) to generate and interpret molecular-scale signals. Seeking Scientific ML / Applied Scientist roles in biophysics and drug discovery.

TECHNICAL SKILLS

- Programming: Python; Bash; Git.
- Machine learning: XGBoost; SVM; scikit-learn; supervised learning; evaluation; feature engineering
- Molecular simulation: AMOEBA, CHARMM; Tinker, GROMACS; MDAnalysis.
- Quantum chemistry: VASP; Gaussian.

EDUCATION

Virginia Tech — Ph.D. in Chemistry, GPA 3.85 (Aug 2022–Present)

Advisor: Valerie Welborn

University of Oregon — M.S. in Chemistry, GPA 3.38 (Sep 2019–Sep 2020)

China University of Mining and Technology (CUMT) — B.S. in Applied Chemistry, GPA 3.67 (Sep 2015–Jun 2019)

SELECTED PROJECT

Binding Pocket Prediction from Protein Structure using XGBoost Classifier (Advanced ML Course Project)

- Implemented a grid-based pocket detection pipeline and trained an XGBoost classifier using geometric and pharmacophore features from local protein neighborhoods.
- Labeled pocket regions from known ligand coordinates (≤ 4 Å) and evaluated on held-out data with standard classification metrics; achieved $F1 \approx 0.92$.
- Scaled inference with cKDTree spatial indexing and batch processing for large 3D grids; exported probability-scored pocket points as PDB files for visualization.

Electrolyte Selection for Sodium Ion Batteries (May 2024–Oct 2024; manuscript in preparation)

- Simulated sodium ion electrolytes in diglyme (0.6 M NaPF₆, 1 M NaPF₆, 1 M NaOTf) using AMOEBA and CHARMM force fields.
- Quantified sodium ion transport and solvation structure (e.g., RDF/diffusion) to rationalize experimental trends and guide electrolyte selection.

Peptide Hydration Water Dynamics in Glucose Solution (Apr 2025–Present)

- Built a benchmark dataset of solvated amino acid pentamers and sequence mixtures across 0–2 M glucose to quantify composition-dependent hydration dynamics.
- Computed hydration and conformational metrics, including velocity autocorrelation function (VACF) fits for diffusion and retardation, H-bond statistics, RDF, Radius of Gyration, and solvent-accessible surface area (SASA) to characterize glucose-induced dynamical shifts.

Glycosylated Enzyme to Reduce Product Inhibition in DszB (Dec 2022–May 2025; published in Protein Science)

- Artificially glycosylated DszB with β -glucose and predicted loop-dynamics changes using AMOEBA MD simulations.
- Computed structural/dynamical metrics (RMSD/RMSF, Ramachandran analysis) to connect glycation to reduced product trapping; published in Protein Science.

PUBLICATIONS

- Liang, J., Zheng, Y., & Vaissier Welborn, V. Mitigating product inhibition in 2'-hydroxybiphenyl-2-sulfinate desulfinate (DszB) with synthetic glycosylation. Protein Science (2025).